

OPERATING SYSTEMS

Memory Management

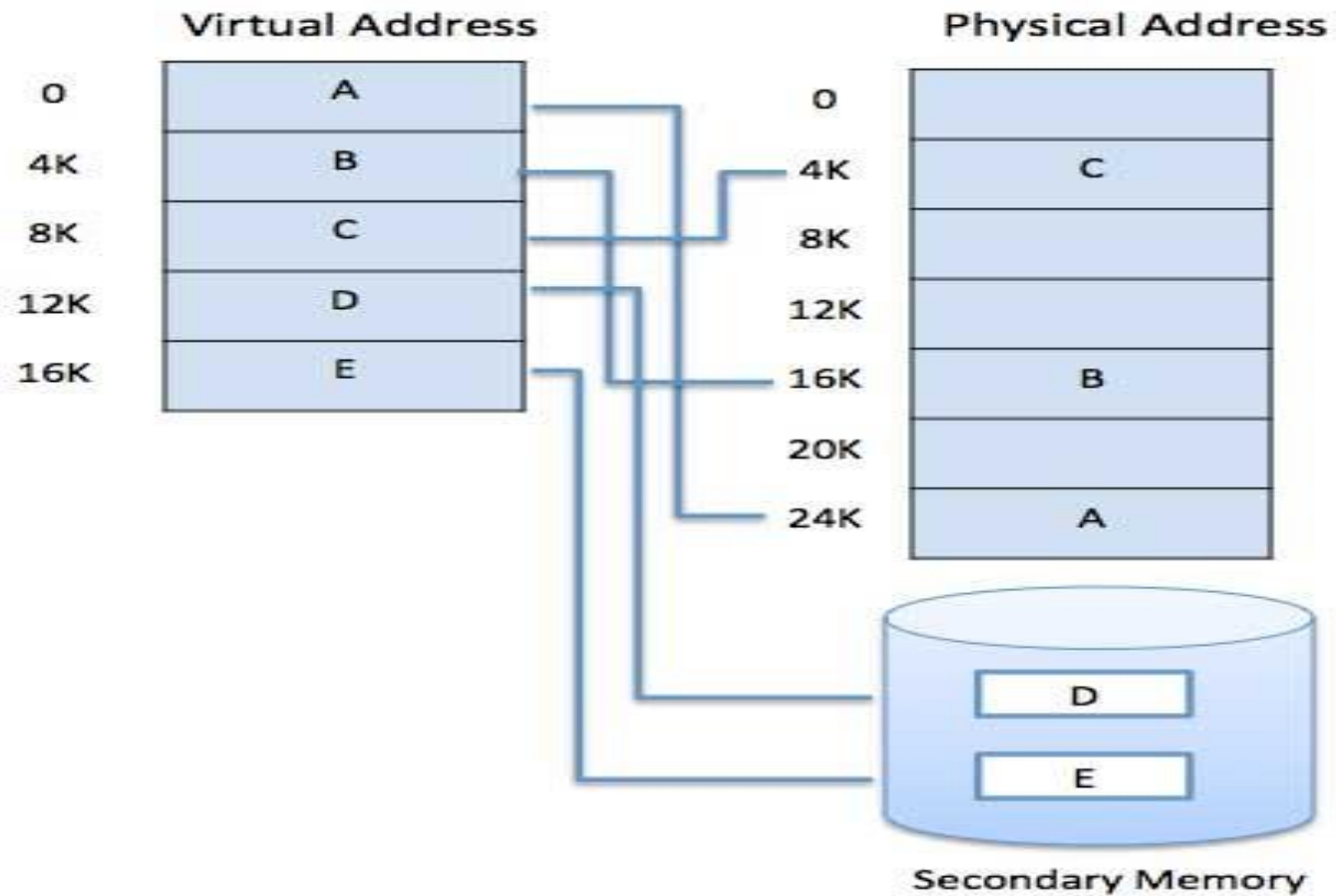
S.Thenmozhi

Department of Computer Applications

- *Virtual memory* is imaginary memory: it gives you the illusion of a memory arrangement that's not physically there.
- A computer can address more memory than the amount physically installed on the system.
- This extra memory is actually called **virtual memory** and it is a section of a hard disk that's set up to emulate the computer's RAM
- The main advantage of this scheme is that programs can be larger than physical memory.
- Virtual memory can be implemented via:
 - Demand paging
 - Demand segmentation

OPERATING SYSTEMS

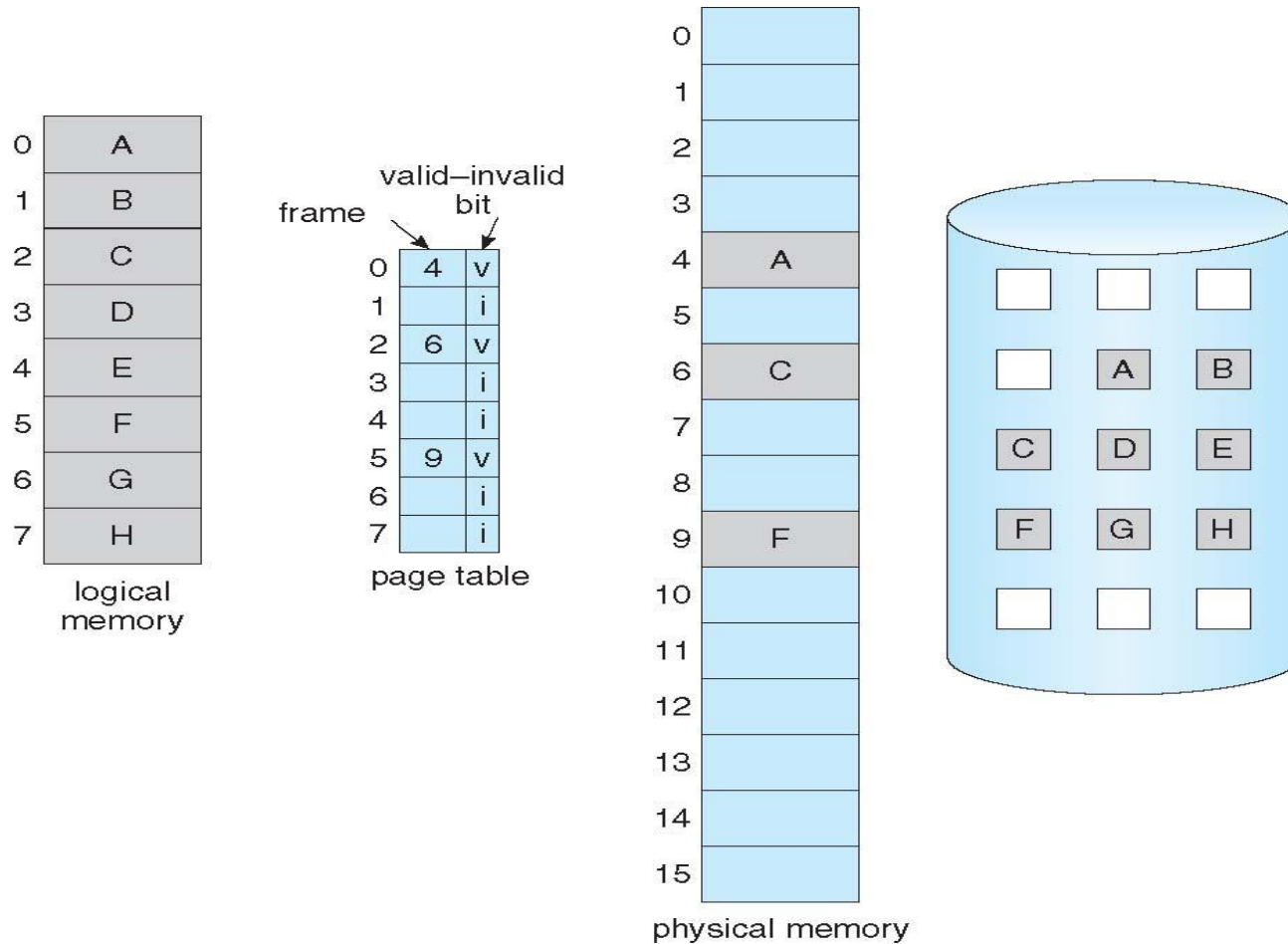
Virtual Memory



- Bring a page into memory only when it is needed
 - Less I/O needed, no unnecessary I/O
 - Less memory needed
 - Faster response
 - More users
- **Lazy swapper** – never swaps a page into memory unless page will be needed
- Page is needed $\Rightarrow$ reference to it
 - invalid reference $\Rightarrow$ abort
 - not-in-memory $\Rightarrow$ bring to memory

- With each page table entry a valid–invalid bit is associated
(**v** $\Rightarrow$ in-memory – **memory resident**, **i** $\Rightarrow$ not-in-memory)
- Initially valid–invalid bit is set to **i** on all entries
- During address translation, if valid–invalid bit in page table entry
is **i** $\Rightarrow$ page fault

Page tables not in memory



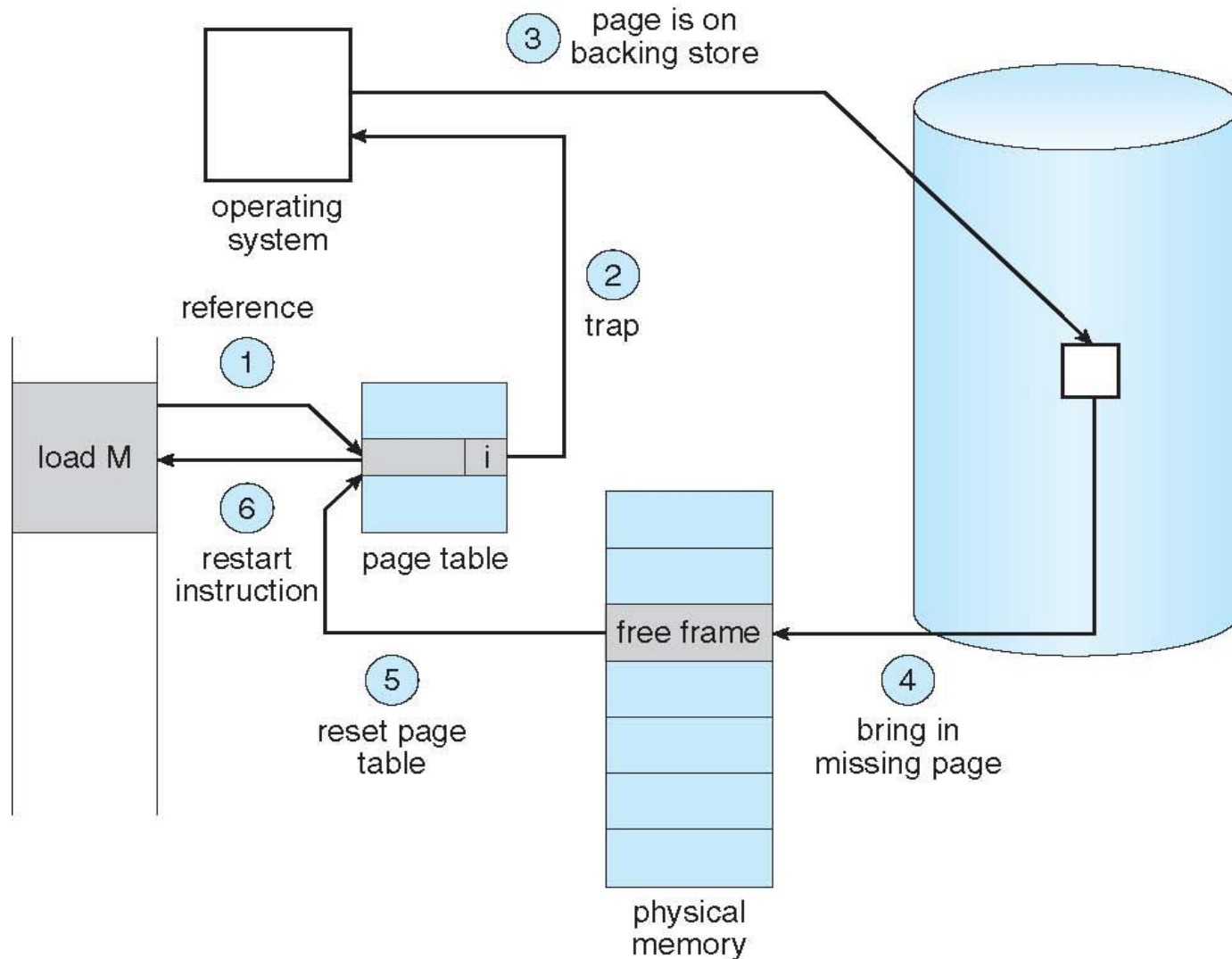
- If there is a reference to a page, first reference to that page will trap to operating system:

page fault

1. Operating system looks at another table to decide:
 - Invalid reference $\Rightarrow$ abort
 - Just not in memory
2. Get empty frame
3. Swap page into frame via scheduled disk operation
4. Reset tables to indicate page now in memory
Set validation bit = **v**
5. Restart the instruction that caused the page fault

OPERATING SYSTEMS

Demand Paging



- Extreme case – start process with *no* pages in memory - **Pure demand paging**
- Actually, a given instruction could access multiple pages -> multiple page faults -> Paging to done based on **locality of reference**
- Hardware support needed for demand paging
 - Page table with valid / invalid bit
 - Secondary memory (swap device with **swap space**)
 - Instruction restart



THANK YOU

S. Thenmozhi

Department of Computer Applications

thenmozhis@pes.edu

+91 80 6666 3333 Extn 393